Effect of Centralized School Choice on College Admissions Outcomes

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Abstract

Centralized school choice systems are increasingly adopted worldwide, yet their longer-term consequences for students' educational trajectories remain poorly understood. This paper examines the causal effect of implementing a centralized school assignment system in secondary education on subsequent college admissions outcomes. We exploit the staggered, region-by-region rollout of Chile's centralized school choice system, which provides quasi-experimental variation in exposure across cohorts, and estimate dynamic treatment effects to accommodate variation in treatment timing. Linking administrative records from secondary and postsecondary education, we find that centralized school choice significantly increases the likelihood that students apply to, are admitted to, and enroll in programs within the centralized college admissions system. These gains are concentrated among female students, students from socioeconomically vulnerable schools, and students from lower-performing schools. We further show that the improvements are driven by admissions to less selective and historically undersubscribed programs, with no evidence of displacement in highly selective programs, alleviating concerns about congestion effects. The results are robust across specifications, placebo tests, and sample restrictions. Together, our findings demonstrate that centralized school choice can meaningfully expand access to higher education, particularly for students historically underserved by decentralized admissions.

1 Introduction

Centralized school choice systems have become a defining feature of modern education policy, with nearly 40% of countries relying on centralized clearinghouses to assign students to secondary schools Neilson (2024). These systems are designed to reduce discriminatory admissions practices, lower application costs, improve information provision, and replace fragmented school-level procedures with a transparent and standardized mechanism. By eliminating uncoordinated offers, early contracting, and inefficient reassignment of seats, centralized assignment promises both greater fairness and improved allocative efficiency.

Despite these advantages, the adoption of centralized assignment has been politically contentious. In Chile, for example, opponents argued that the centralized system would undermine parental autonomy and restrict families' ability to choose their children's schools (Rebolledo and Olea 2018). Similar debates have emerged in other countries, where concerns about bureaucratic overreach, loss of control, or reduced flexibility have shaped public opinion. Beyond these political dynamics, the expected effects of centralized school choice on downstream outcomes such as college applications, admissions, and enrollment are theoretically ambiguous. Eliminating selective screening at the school level may improve academic preparation and information for disadvantaged students, but it may also weaken school–student matching or reduce incentives for effort.

Studying the effects of centralized school choice—and the mechanisms through which these reforms operate—is empirically challenging. Reforms of this type are typically implemented once, at a local level, and without natural comparison groups, making it difficult to construct valid counterfactuals and assess their broader implications. Moreover, centralized assignment often coincides with other policy changes—such as funding reforms, accountability shifts, or curricular adjustments—that further complicate identification. As a result, the existing literature has largely focused on district-level reforms to analyze short-run effects on sorting, access, or satisfaction, leaving open the question of how centralized school choice shapes students' postsecondary trajectories.

In this paper, we address these challenges by exploiting the staggered implementation of Chile's centralized school choice system (SAE), which was rolled out region by region between 2016 and 2020 following the Ministry of Education's pseudo-random selection of regions for early adoption. This staggered rollout provides a unique opportunity to study the causal effect of centralized school choice on college admissions outcomes because it creates quasi-experimental variation in exposure across regions and cohorts. Regions that have not yet implemented the reform provide a natural comparison group for those that have already adopted it, helping disentangle the effect of the policy from broader trends affecting students' higher-education trajectories. The setting is particularly well suited for this analysis because Chile's college admissions system is itself centralized for the most selective institutions, allowing us to trace how changes in secondary-school assignment propagate into higher-education choices.

Our findings reveal four main results. First, centralized school choice significantly increases the likelihood that students apply to, are admitted to, and enroll in programs within the centralized college admissions system. Second, these effects are heterogeneous: the gains are concentrated among female students, students from high-vulnerability schools, and students from lower-performing schools. Third, the improvements are driven primarily by admissions to less selective and historically under-subscribed programs, with no evidence of displacement in highly selective programs, alleviating concerns about congestion effects. Finally, the results are robust across alternative specifications, placebo tests, parallel-trends diagnostics, and sample restrictions. Together, these findings show that centralized school choice can meaningfully expand access to higher education, particularly for students historically underserved by decentralized admissions.

The remainder of the paper is organized as follows. Section 2 discusses the most closely related literature. Section 3 provides institutional background on Chile's school and college admissions systems. Section 4 presents the main results. Section 5 concludes.

2 Literature

Our work relates to three main strands of the literature.

2.1 School Choice and Student Outcomes

A large empirical literature examines how expanding school choice—i.e., relaxing traditional neighborhood-based assignment policies and allow families to select among multiple schools, including magnet and voucher programs, open-enrollment initiatives, and inter-district choice—affects student outcomes. Early work focused on cross-district or cross-municipality variation in competitive pressure, with mixed findings on achievement, school quality and productivity (Hoxby 2000, 2003, Hsieh and Urquiola 2006, Rothstein 2007). Other studies exploit lotteries or oversubscription rules to estimate the individual effects of attending preferred schools (Cullen et al. 2006, Abdulkadiroğlu et al. 2014, Deming et al. 2014, Muralidharan and Sundararaman 2015). These papers typically evaluate short-run impacts on test scores, peer composition, or satisfaction, and do not study how centralized assignment shapes longer-term educational trajectories. The study most closely related to ours is Campos and Kearns (2024), who analyze Los Angeles's Zones of Choice (ZOC), a district-level reform that created small local choice markets while retaining neighborhood boundaries elsewhere. The authors show that student outcomes in ZOC markets increased markedly, narrowing achievement and college enrollment gaps between ZOC neighborhoods and the rest of the district. Furthermore, the authors identify competition-induced improvements in school quality as the central mechanism behind the effects of the reform.

2.2 Centralized School Choice.

A parallel literature studies the design and consequences of *centralized* school choice systems, in which families submit rank-ordered preference lists and a central clearinghouse assigns students to schools using a matching algorithm. Starting with Abdulkadiroğlu and Sönmez (2003), who formalize school choice as a mechanism design problem and compare the properties of the two most widely used assignment rules—*deferred acceptance* and *immediate acceptance*—this literature has developed along several complementary lines. First, a large theoretical literature studies the incentive, efficiency, and stability properties of alternative matching mechanisms and the strategic behavior they induce (Abdulkadiroğlu and Sönmez 2003, Kesten 2010, Pathak 2017). Second, a set of papers evaluates the implementation of centralized assignment systems in large urban school districts, documenting how changes in the mechanism affect equilibrium behavior, strategic reporting of preferences, and assignment outcomes (Abdulkadiroğlu et al. 2005, 2009). Third, a growing empirical literature uses administrative data from centralized clearinghouses to estimate students' preferences over schools and to evaluate welfare and distributional consequences of school choice systems (Hastings et al. 2008, Agarwal and Somaini 2018). Finally, recent work studies how centralized assignment reforms affect broader outcomes such as access to high-quality schools, school segregation, and students' educational trajectories (Neilson 2021).

2.3 Centralized College Admissions.

Finally, our work is also related to a literature studying the drivers of outcomes in *centralized college admissions systems*. One strand of this literature focuses on the *design of the assignment mechanism*. In particular, this work studies how different elements of the allocation process—including the algorithm used (Calsamiglia et al. 2020), exam-retaking and re-application policies (Larroucau and Rios 2025, Rios and Sharma 2026), tie-breaking rules (Abdulkadiroğlu et al. 2009, Arnosti 2015, Ashlagi et al. 2019), affirmative action policies (Rios et al. 2021a, Baswana et al. 2019), the timing of allocations (Luflade 2017, Gong and Liang 2022), and the aftermarket (Kapor et al. 2024)—affect the theoretical properties of the mechanism, students' strategic behavior, and equilibrium allocation outcomes (Chen and Kesten 2013, Fu 2014, Neilson 2021). Another strand studies the role of *information frictions and beliefs* in shaping students' application behavior. In many cases, students lack complete information about program characteristics, hold biased beliefs about their admission probabilities, or are unaware of alternative programs that may be a good fit. As a result, they may submit suboptimal preference lists, which can be corrected through information policies implemented at scale (Hastings et al. 2008, Agarwal and Somaini 2018, Larroucau et al. 2024).

2.4 Contribution

Our work contributes to and extends these literatures in two important ways. First, much of the existing empirical work on school choice evaluates policy changes implemented at the district or local level that expand families' ability to choose among schools. In contrast, the reform we study represents a nationwide institutional transformation: the introduction of the *Sistema de Admisión Escolar* (SAE), which replaced decentralized, school-level admissions processes with a unified centralized clearinghouse based on a deferred-acceptance algorithm. This reform eliminated discretionary selection practices by schools and standardized admissions rules across all publicly funded institutions. As a result, it provides a unique setting to study how a large-scale transition to centralized assignment affects student behavior and outcomes.

Second, our paper examines a different set of outcomes than most of the existing literature. While prior work on school choice and centralized assignment primarily focuses on short-run outcomes such as test scores, peer composition, or access to preferred schools, our data allow us to link administrative records across secondary and postsecondary education. This linkage enables us to study how exposure to centralized school assignment during secondary education affects students' subsequent participation and outcomes in a separate nationwide centralized college admissions system.

More broadly, the literature on centralized college admissions emphasizes that outcomes depend critically on institutional design and students' application behavior within the system. In contrast, our analysis highlights how experiences *prior* to college admissions can shape behavior in centralized matching environments. Specifically, we study whether participation in a centralized school choice clearinghouse during secondary education affects students' application, admission, and enrollment outcomes when they later participate in centralized college admissions. By linking these two centralized markets, our paper shows how earlier exposure to centralized choice institutions can influence behavior and outcomes in subsequent matching environments.

3 Background

3.1 School Choice.

Before the reform, admissions to Chile's public and publicly subsidized (voucher) schools were conducted through fully decentralized processes. Each school operated its own admissions procedure, often relying on discretionary and exclusionary practices such as interviews with parents and students, unofficial entrance exams, and reviews of past academic records. Because offers were not coordinated, families frequently faced pressure to accept early admissions without knowing whether more preferred schools would later admit them, and declined seats were rarely efficiently reassigned. Many schools also used first-come, first-served queues, leading parents to wait overnight to secure a seat (Correa et al. 2022). These features contributed to high levels of socioeconomic segregation in the Chilean school system.

In 2015, the Chilean Congress enacted the *School Inclusion Law*, which fundamentally restructured school admissions. The law eliminated copayments in publicly subsidized schools, prohibited any form of selection based on academic, social, religious, or economic criteria, and mandated the use of a centralized assignment system. The law also established three priority groups—siblings of current students, children of school employees, and returning students—that must be served in strict order when seats are oversubscribed, with ties within each group broken using a random tie-breaker (Correa et al. 2022). In addition, it introduced quotas for students with special educational needs, high academic performers, and disadvantaged students. This newly established centralized system, which covers all grades from prekindergarten through 12th across all regions, collects families' ranked preferences through an online platform, provides standardized information about schools, and assigns each student to at most one school using a variant of the deferred acceptance (DA) algorithm (Correa et al. 2022).

3.2 College Admissions.

Chile's college admissions system is semi–centralized, with the most selective universities participating in a single centralized admissions system and the remaining institutions conducting their admissions independently.

Applicants are evaluated solely based on scores that come from (i) a national entrance exam, which includes mandatory Mathematics and Language components and optional subject tests in Sciences or History; and (ii) their high-school performance, captured by a grade-based score (NEM) and a rank-based score reflecting a student's standing within their graduating cohort. Knowing their scores, students apply through a centralized online platform by submitting a rank-ordered list of up to ten university–major programs. After applications are submitted, the clearinghouse verifies eligibility requirements, computes each applicant's program-specific scores based on weights previously announced by each university, and ranks applicants accordingly. Final assignments are produced using a variant of DA that assigns each applicant to the highest-ranked program on their list for which they qualify and have an application score above the program's clearing cutoff (Rios et al. 2021b). Once results are released, students may choose whether to enroll in the program to which they were assigned or instead matriculate outside the centralized system.

3.3 Data.

A distinctive feature of our dataset is that it allows us to follow each student continuously from secondary school into higher education, enabling the construction of outcomes that capture the full academic pipeline.

Secondary education. Our dataset includes rich administrative records covering the full secondary-school trajectory of each student. For every school-year, we observe enrollment, attendance, graduation, and level grades. Once the region became part of the centralized school assignment system (SAE), we additionally observe families' reported preferences over schools, school capacities, the priority-based orderings used to process applications (combining priority groups with random tie-breakers), and the resulting assignment outcomes. These data allow us to reconstruct each student's academic history from the beginning of secondary education through completion of 12th grade.

Tertiary education. We complement these records with comprehensive information on students' postsecondary trajectories. For every year, we observe enrollment and graduation outcomes in any higher-education institution, regardless of whether the institution participates in the centralized admissions system. For students who do participate in the centralized process, we additionally observe their standardized test scores, high-school–based admission scores, rank-ordered program applications, and program-level admission results. This combination of centralized and non-centralized data provides a complete view of students' opportunities and choices in the tertiary sector.

Outcomes. Because our empirical strategy aggregates these individual records to the school-year level, we compute a set of college-admissions-related outcomes for each school and cohort. These include the school's average high-school GPA, average Math–Verbal test score, the share of students who apply to the centralized system, the share who receive an assignment, the share who enroll in a program within the centralized system, and the share who enroll in any postsecondary program outside the centralized system. Together, these outcomes summarize both academic preparation and postsecondary decision-making.

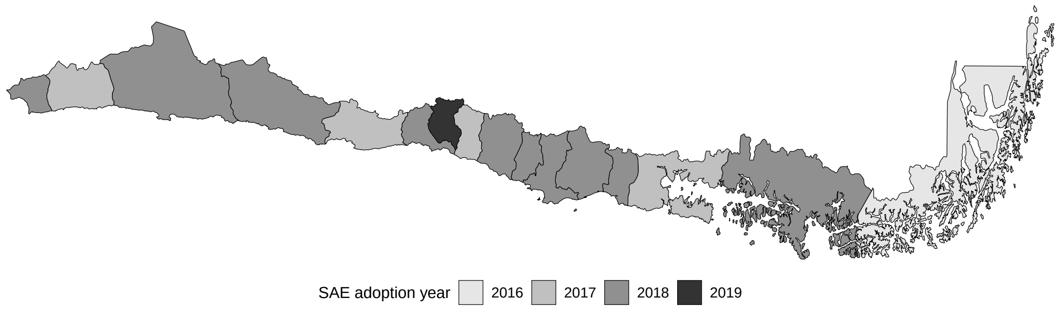

Figure 1: Implementation Year for 9th-grade

Sample. Our analysis focuses on students who completed all four years of secondary education (grades 9–12) in the same institution and on time. Restricting the sample in this way avoids conflating treatment effects with mobility across schools or grade repetition. Furthermore, we begin following students in 9th grade since this is the typical entry point into secondary education and the grade at which most students would have been admitted when the centralized school choice system (SAE) was introduced.

Unless otherwise stated, we restrict attention to cohorts that entered 9th grade between 2012 and 2021. Focusing on cohorts that entered 9th grade in 2012 ensures that, by the time they applied to college, the nationwide free-tuition policy—which first applied to students enrolling in 2016—was already fully in place, eliminating concerns that differential exposure to the reform could drive our results. At the same time, restricting the sample to cohorts up to 2021 ensures that these students participated in the 2025 college admissions process—the last year for which we have data—allowing us to observe and measure their postsecondary outcomes.

3.4 Empirical Strategy

The introduction of the centralized school choice system in Chile provides a natural setting to identify the causal effects of centralized assignment on school-level outcomes. Beginning in 2016, the Ministry of Education implemented the centralized school choice system (SAE) through a phased regional rollout that was planned to take four years. The Magallanes region was selected as the initial pilot in 2016, while the Metropolitan Region—given its size—was incorporated in the final stage of the expansion. Among the remaining regions, adoption was effectively randomized: half entered the system in 2017 and the other half in 2018. Importantly, the timing of adoption was not announced in advance, limiting the scope for parents to adjust their behavior in anticipation. Figure 3.4 reports the implementation year for 9th-grade admissions in each region.

The combination of (i) staggered rollout across regions, (ii) partially random and unanticipated assignment of regions to early versus late adoption cohorts, and (iii) the exclusion of private schools from the centralized system—creating a clear treated–control comparison across school types within regions and over time—yields an ideal environment for a staggered difference-in-differences (DiD) design. Specifically, we estimate dynamic treatment effects under staggered adoption using the framework of Callaway and Sant'Anna (2021). Let $Y_{it}(0)$ and $Y_{it}(1)$ denote the potential outcomes for unit i at time t in the absence and presence of treatment, and let $G_i = g$ indicate that unit i first becomes treated in period g. For each cohort g and time $t \geq g$, the causal parameter of interest is the group–time average treatment effect,

$$ATT(g,t) = \mathbb{E}[\, Y_{it}(1) - Y_{it}(0) \mid G_i = g\,]\,.$$

Based on these estimates, the overall effect of the policy change can be estimated using simple aggregated ATT that captures the weighted average of all group-time average treatment effects with weights proportional to the group size, i.e.,

$$A\bar{T}T = \sum_{g \in G} \sum_{t=0}^{T} ATT(g,t) \cdot \omega(g,t)$$

where $\omega(g,t)$ is the proportion of observations coming from cohort g and period t. Unless otherwise stated, the estimates reported in the reminder of the paper are simple aggregated ATTs.

4 Results

Table 1 reports our main results. In each case, we control for the number of students in the cohort and report double robust standard errors adjusted for multiple hypothesis testing.

Table 1: Regression Results

	GPA	Average LM	Applied	Assigned	Enrolled
	(1)	(2)	(3)	(4)	(5)
Treated	0.001	-0.002	0.055***	0.046***	0.035***
	(0.006)	(0.003)	(0.005)	(0.005)	(0.005)
Mean (pre-treatment)	5.651	0.416	0.412	0.306	0.247
Observations	34,318	34,318	34,318	34,318	34,318

We observe that the average impact of the policy is positive and significant for several outcomes. Students who graduated from a treated school after the implementation of the policy were 13.35%

more likely to apply to the centralized college admissions system relative to their average in the pre-treatment period. Similarly, treated students where 15.03% more likely to be assigned and 14.17% more likely to enroll in a program that is part of the centralized college admissions system. Furthermore, we observe no significant effect on average high-school GPA nor in the average Math-Verbal score obtained in the entrance exam.

4.1 Heterogeneity

4.1.1 Gender.

Table 2 reports the estimated aggregated treatment effects for each outcome of interest, separately by gender. Consistent with Table 1, we find no effect on high-school GPA or on students' average Math–Verbal score in the entrance exam. By contrast, we find positive and significant effects on the remaining outcomes for both genders, with the effect on female students being specially strong: their probability of applying to the centralized admissions system increases by 17.73%, the probability of being admitted to at least one program in the system rises by 23.63%, and the probability of enrolling in a program within the system increases by 17.79%.

Table 2: Effect on Outcomes by Gender

	GPA		Average LM		Applied		Assigned		Enrolled	
	Female	Male	Female	Male	Female	Male	Female	Male	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treated	0.002	0.001	-0.001	-0.004	0.072***	0.037***	0.054***	0.040***	0.043***	0.026***
	(0.007)	(0.007)	(0.003)	(0.004)	(0.006)	(0.006)	(0.007)	(0.006)	(0.006)	(0.006)
Mean pre-treatment	5.710	5.573	0.418	0.412	0.434	0.382	0.307	0.302	0.243	0.251
Observations	33,446	32,479	33,446	32,479	33,446	32,479	33,446	32,479	33,446	32,479

4.1.2 School Vulnerability.

The most commonly used measure of schools' socioeconomic vulnerability in Chile is the *Índice de Vulnerabilidad Escolar (IVE-SINAE)*, which measures the fraction of students enrolled in a school who are classified as socioeconomically vulnerable. The classification relies on administrative records and survey data on students' household conditions, including income, parental education, employment status, housing conditions, and participation in social assistance programs.

To examine whether the reform's effects vary with schools' socioeconomic vulnerability, we split treated schools into two groups based on their IVE-SINAE: *low vulnerability* (below the median) and *high vulnerability* (above the median). Because never-treated schools (i.e., private schools) do not report this metric, we estimate the average treatment effect separately for low- and high-vulnerability schools, and in both cases include private schools in the sample.

Table 3 shows that students in both low- and high-vulnerability schools experience significant increases in application, admission, and enrollment rates following the transition from decentralized to centralized school choice. The magnitude of these gains, however, is substantially larger in high-vulnerability schools: application rates rise by 25.29% compared with 6.89% in low-vulnerability schools; admission rates increase by 33.73% versus 6.77%; and enrollment rates by 34.81% versus 5.29%. These patterns indicate that the reform significantly expanded access for students in more socioeconomically disadvantaged schools.

Table 3: Effect on Outcomes by Vulnerability

	GPA		Average LM		Applied		Assigned		Enrolled	
	Low	High	Low	High	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treated	0.012*	-0.009	-0.002	-0.004	0.039***	0.065***	0.030***	0.057***	0.019***	0.047***
	(0.006)	(0.007)	(0.003)	(0.004)	(0.009)	(0.006)	(0.010)	(0.005)	(0.007)	(0.005)
Mean pre-treatment	5.729	5.569	0.447	0.384	0.566	0.257	0.443	0.169	0.359	0.135
Observations	20,904	20,918	20,904	20,918	20,904	20,918	20,904	20,918	20,904	20,918

4.1.3 School Performance.

Students in 10th grade take a nationwide standardized exam—the *Sistema de Medición de la Calidad de la Educación* (SIMCE)—which assesses mastery of the official curriculum. SIMCE scores are widely used as a proxy for school quality. To examine whether the reform's effects vary with school quality, we use schools' SIMCE performance in 2015—specifically, the average of the Math and Verbal components for 10th-grade students—and classify schools that eventually become treated into two groups: (i) *high-quality* schools, with an average score above the median, and (ii) *low-quality* schools, with an average score below the median.

Table 4 reports the estimated aggregated treatment effects for each outcome of interest, separately by school quality. The positive effects on applications, admissions, and enrollment are primarily driven by students from low-quality schools. For these outcomes, the aggregated average treatment effect is positive and statistically significant, with application, assignment, and enrollment rates increasing by 25%, 28.72%, and 33.33% relative to the pre-treatment period, respectively. In contrast, we find no statistically significant effects among students from high-performing schools.

Table 4: Effect on Outcomes by School Quality

	GPA		Average LM		Applied		Assigned		Enrolled	
	Low	High	Low	High	Low	High	Low	High	Low	High
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Treated	-0.107**	0.005	-0.040	-0.006*	0.072***	0.001	0.056**	0.002	0.052**	0.007
	(0.043)	(0.008)	(0.049)	(0.004)	(0.024)	(0.007)	(0.025)	(0.007)	(0.022)	(0.007)
Mean pre-treatment	5.583	5.727	0.392	0.442	0.288	0.550	0.195	0.431	0.156	0.349
Observations	21,080	13,238	21,080	13,238	21,080	13,238	21,080	13,238	21,080	13,238

4.1.4 Program Selectivity.

The results in Table 1 indicate that transitioning from decentralized to centralized admissions increased the probability of being assigned to a program within the centralized college admissions system. However, programs differ substantially in their quality and selectivity: some are highly competitive, while others routinely fail to fill their regular seats. Identifying which types of programs drive the overall improvement is essential for assessing potential congestion effects that could bias the estimated impact of the reform. In particular, if the reform disproportionately benefits students from schools that joined SAE—especially when they apply to selective programs—these students may displace applicants who did not experience the policy change. Such displacement would generate a congestion effect that could upwardly bias the estimated treatment effect.

To investigate this possibility, Table 5 reports the estimated aggregated treatment effects on the probability of admission to a program within the centralized system, disaggregated by whether the program filled its seats and by its selectivity level, with the latter measured using the program's cutoff position in the distribution of regular-process cutoffs (10th, 25th, 50th, 75th, and 90th percentiles).[1]

We find no effect on programs that fill their seats, but we observe a positive and significant effect among programs that do not. Consistent with this pattern, we detect no significant effect on admission to programs in the top 25th percentile of the selectivity distribution and even a negative effect at the top 10th percentile. In contrast, we find positive and significant effects on admission to less selective programs—those at the 50th, 75th, and 90th percentiles.

[1] Only 53% of programs in the centralized system filled their seats in 2025.

Table 5: Regression Results

	All	Full	Not Full	Top 10	Top 25	Top 50	Top 75	Top 90
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Treated	0.046***	0.001	0.045***	-0.022***	-0.003	0.036***	0.046***	0.046***
	(0.005)	(0.004)	(0.003)	(0.003)	(0.003)	(0.005)	(0.005)	(0.005)
Mean pre-treatment	0.306	0.234	0.073	0.041	0.100	0.221	0.301	0.306
Observations	34,318	34,318	34,318	34,318	34,318	34,318	34,318	34,318

Taken together, these results indicate that the overall improvement in assignment probabilities is driven by increased admissions to less selective programs that routinely operate below capacity. This pattern suggests that the reform does not generate meaningful congestion in highly selective programs and is therefore unlikely to induce substantial congestion-related bias in the estimated treatment effects, consistent with other interventions in the Chilean higher education system (Larroucau and Rios 2025).

4.2 Robustness Checks

This section provides a series of robustness checks that validate and strengthen the results discussed above.

4.2.1 Parallel Trends.

Critical to the validity of our empirical strategy is a cohort-specific parallel trends assumption, which states that, in the absence of treatment, the evolution of potential outcomes for cohort g would have mirrored that of units that remain untreated in period t.[2]

To test this assumption, we follow Callaway and Sant'Anna (2021) and estimate event-time average treatment effects for each relative period $e = t - g$, where g denotes the cohort's first treatment period. For each event time, the estimator identifies

$$ATT(e) = \mathbb{E}[\,Y_{it}(1) - Y_{it}(0) \mid t - g = e\,]$$

[2]In implementing the parallel trends test, we use units that are *not yet treated* as the comparison group rather than units that are never treated. As emphasized by Callaway and Sant'Anna (2021), never-treated units need not provide a valid counterfactual for treated cohorts when they differ systematically in observable characteristics. In our setting, never-treated students attend private schools and have substantially higher-income, making them unsuitable as a comparison group for assessing pre-treatment trends. By contrast, not-yet-treated units are drawn from the same population of eventually treated cohorts and therefore provide a more credible basis for evaluating whether treated and comparison units followed similar trends prior to treatment.

by comparing treated units to those that are yet-to-be-treated in period t. Under the cohort-specific parallel trends assumption, all pre-treatment effects must be zero, implying the joint null hypothesis $ATT(e) = 0$ for all $e < 0$.

Table 6 reports the estimated event-time average treatment effects for each relative period across all outcomes. Most of the pre-treatment coefficients are small and statistically indistinguishable from zero. The only exceptions occur at event times -7 and -1 for the assignment outcome. In contrast, after the policy change, we observe positive and statistically significant effects for most post-treatment periods. Taken together, these results are consistent with the parallel trends assumption in the pre-treatment period.

4.2.2 Placebo Tests.

Private Schools. A key feature of our setting is the presence of a group of never-takers. Since SAE only includes publicly funded schools (i.e., public and voucher), private schools are never exposed to the system. If congestion or spillover effects generated by the reform are limited, outcomes for private schools should remain unchanged when their region joins the centralized system, even though these schools themselves do not participate. In other words, treating regional adoption as a "placebo treatment" for private schools should yield no detectable effects.

We implement this placebo by estimating the simple aggregated ATT restricting the sample to private schools only, assuming as treatment period the one when SAE was implemented in the school's region. Table 7 shows that none of the estimated coefficients is statistically significant. This pattern indicates that private-school outcomes did not change when public and voucher schools in their region entered SAE, consistent with the absence of meaningful spillover or congestion effects operating through the centralized admissions market.

Timing. Another key feature of our setting is the staggered rollout of SAE across regions, which allows us to construct a placebo–timing test by artificially shifting each region's adoption year to an earlier period. To implement this, we assign a placebo implementation year that precedes the true one by a fixed number of years and restrict the sample to periods strictly before the actual adoption date, ensuring that all observations remain untreated. We then re-estimate the $ATT(t, g)$ for each placebo shift. If the identifying assumptions hold and there are no differential pre-trends, these placebo treatments should yield no significant effects—or effects that are much smaller than those obtained using the true implementation year.

For each outcome, Figure 2 displays boxplots with the estimates obtained considering different

Table 6: Parallel Trends Assumption: Event-Study Average ATT

	GPA	Average LM	Applied	Assigned	Enrolled
	(1)	(2)	(3)	(4)	(5)
Event time -9	0.039	0.008	0.003	-0.008	0.000
	(0.028)	(0.023)	(0.021)	(0.027)	(0.025)
Event time -8	0.001	0.009	-0.006	-0.002	-0.003
	(0.006)	(0.004)	(0.006)	(0.005)	(0.005)
Event time -7	0.001	-0.002	**0.016**	**0.013**	**0.012**
	(0.005)	(0.003)	(0.004)	(0.004)	(0.004)
Event time -6	-0.006	0.002	-0.003	-0.003	0.001
	(0.004)	(0.003)	(0.004)	(0.003)	(0.003)
Event time -5	-0.001	-0.003	0.010	0.010	0.008
	(0.005)	(0.003)	(0.004)	(0.004)	(0.003)
Event time -4	0.005	0.002	0.009	0.005	0.006
	(0.004)	(0.003)	(0.004)	(0.004)	(0.003)
Event time -3	0.001	-0.004	-0.008	-0.009	-0.003
	(0.005)	(0.003)	(0.004)	(0.003)	(0.003)
Event time -2	-0.000	0.001	0.009	0.007	0.006
	(0.005)	(0.003)	(0.004)	(0.004)	(0.004)
Event time -1	-0.007	-0.002	0.002	**0.012**	0.010
	(0.005)	(0.003)	(0.004)	(0.004)	(0.004)
Event time 0	-0.007	0.001	**0.022**	**0.016**	**0.007**
	(0.005)	(0.003)	(0.004)	(0.004)	(0.004)
Event time 1	-0.010	0.001	**0.049**	**0.031**	**0.020**
	(0.006)	(0.003)	(0.005)	(0.005)	(0.005)
Event time 2	0.011	-0.002	**0.065**	**0.055**	**0.040**
	(0.007)	(0.003)	(0.005)	(0.006)	(0.006)
Event time 3	0.015	-0.004	**0.063**	**0.060**	**0.052**
	(0.009)	(0.004)	(0.007)	(0.007)	(0.007)
Event time 4	0.026	-0.003	**0.084**	**0.086**	**0.075**
	(0.013)	(0.006)	(0.010)	(0.010)	(0.010)
Event time 5	-0.044	-0.014	0.053	0.057	**0.137**
	(0.042)	(0.021)	(0.032)	(0.033)	(0.032)

Table 7: Placebo Test: Only Private Schools

	GPA	Average LM	Applied	Assigned	Enrolled
	(1)	(2)	(3)	(4)	(5)
Treated	-0.021	0.002	-0.005	-0.010	-0.013
	(0.016)	(0.009)	(0.013)	(0.013)	(0.013)
Mean (pre-treatment)	6.015	0.462	0.779	0.702	0.608
Observations	4,929	4,929	4,929	4,929	4,929

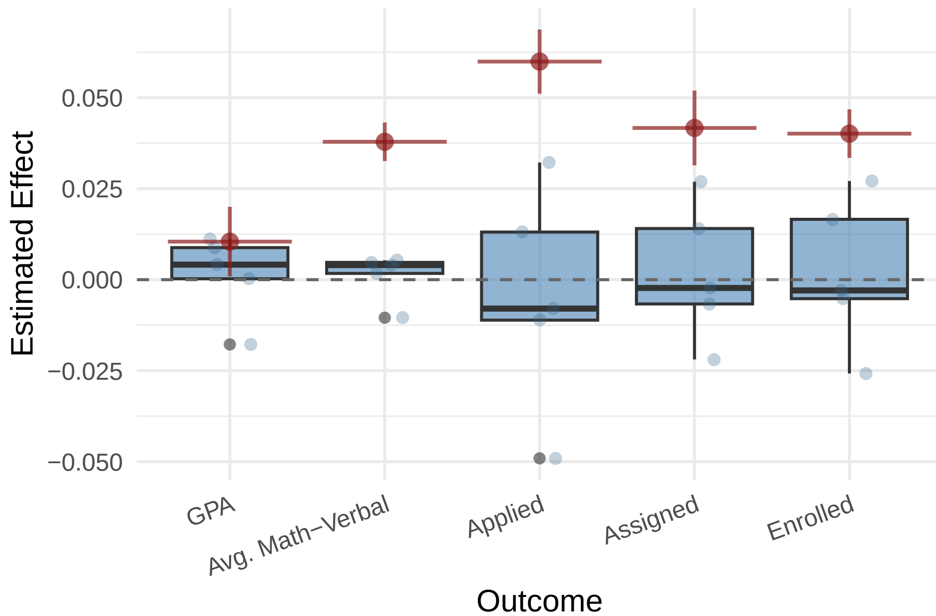

Figure 2: Caption

shifts in the time of implementation of SAE, and also includes (in red) the estimates obtained with the actual year of implementation. We observe that the latter is significantly higher than the former for all the outcomes considered, suggesting that the estimated effects do not arise when the policy is artificially assigned to earlier years. This pattern is consistent with the absence of differential pre-trends and reinforces the interpretation that our main estimates capture the causal impact of SAE rather than spurious pre-existing dynamics.

4.2.3 Two-Way Fixed Effects

A natural approach to evaluate the effect of the reform is to estimate a two-way fixed-effects (TWFE) model of the form:

$$Y_{it} = \alpha_i + \lambda_t + \beta W_{it} + X_{it}\gamma + \varepsilon_{it}, \tag{1}$$

where Y_{it} denotes the outcome for unit i and year t; α_i and λ_t are school and year fixed effects, respectively; X_{it} includes time-varying school characteristics, such as enrollment size; and ε_{it} is an error term. The main variable of interest is W_{it}, which is an indicator equal to one if school i is part of SAE in year t, and zero otherwise.

Although it is intuitive and has been used in prior literature,[3] a growing body of literature shows that this approach is inappropriate with multiple periods and staggered treatment adoption (Borusyak and Jaravel 2018, de Chaisemartin and D'Haultfoeuille 2020, Goodman-Bacon 2021, Sun and Abraham 2021).[4] Nevertheless, Table 8 in Appendix 6.1, which reports the TWFE estimates for completeness, shows results that are broadly consistent with the estimates reported in Table 1.

5 Conclusions

We provide the first nationwide evidence on how transitioning from decentralized to centralized school assignment shapes students' subsequent behavior and outcomes in a separate centralized college admissions process. Leveraging the staggered and partially randomized rollout of Chile's *Sistema de Admisión Escolar* (SAE), we estimate dynamic treatment effects that isolate the causal impact of centralized school choice on the full pipeline from secondary schooling to higher education. Two findings stand out. First, exposure to centralized school assignment substantially increases the likelihood that students apply to, are admitted to, and enroll in programs within the centralized college admissions system. Second, these gains are concentrated among students historically underserved by decentralized admissions—female students, students from socioeconomically vulnerable schools, and students from lower-performing schools—indicating that centralized assignment can meaningfully reduce downstream inequities in access to higher education.

Beyond documenting average effects, our analysis sheds light on the mechanisms through which centralized school choice affects postsecondary trajectories. We show that the improvements are driven by increased admissions to less selective and historically under-subscribed programs, with

[3]For instance, Kutscher et al. (2023) use this strategy to estimate the effect of the reform on the composition of schools.

[4]TWFE constructs its identifying variation by comparing newly treated units not only to never-treated or not-yet-treated units—which provide valid counterfactuals—but also to units that have already been treated. Because already-treated units no longer represent untreated potential outcomes, these comparisons embed treatment effect dynamics and contaminate the estimated coefficient. As a result, the TWFE estimator can combine heterogeneous cohort- and time-specific treatment effects using weights that are difficult to interpret and may even be negative, leading to estimates that do not correspond to any meaningful average treatment effect.

no evidence of displacement in highly selective programs. This pattern suggests that centralized assignment expands participation without generating congestion in the most competitive segments of the market. The absence of effects on high-school GPA and entrance-exam performance further indicates that the reform operates primarily through changes in information, expectations, and application behavior rather than through improvements in academic preparation. Together, these results highlight the importance of institutional design in shaping how students navigate sequential matching environments and demonstrate that early exposure to transparent, standardized assignment processes can influence later decisions in related markets.

From a policy perspective, our findings underscore the broader value of coordinated choice systems in education markets. Centralized assignment not only reduces inefficiencies and inequities in school admissions but also appears to alter students' engagement with subsequent centralized processes, expanding access to higher education for groups that have historically faced barriers in decentralized settings. These insights are particularly relevant for policymakers considering reforms to school or college admissions systems, as they suggest that the benefits of centralized assignment extend beyond the immediate allocation problem and can propagate across educational stages. More broadly, our results point to promising avenues for future work on how early institutional experiences shape behavior in later matching environments, how information and transparency affect students' strategic decisions, and how coordinated systems can be designed to promote equity and efficiency across the educational pipeline.

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6 Appendix

6.1 Additional Results

Table 8: Two-Way Fixed Effects Regression

	GPA	Average LM	Applied	Assigned	Enrolled
	(1)	(2)	(3)	(4)	(5)
Treated	-0.003	-0.002	0.042**	0.042**	0.039**
	(0.007)	(0.002)	(0.017)	(0.016)	(0.016)
Mean (pre-treatment)	5.651	0.416	0.412	0.306	0.247
Observations	34,318	34,318	34,318	34,318	34,318